# *Teaching Notes 1B**
# *Stock Valuation*

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

# Overview

- Returns from Investment in Stocks
- How Do Investors Set The Price Of A Stock?
  - Today: $P_0$ versus Tomorrow: $P_1$
- The Gordon Growth Model (GGM):
  - How Do Financial Analysts Use The GGM?
  - Using The Model To Set Up Prices.
  - How To Estimate The Growth Rate, g.
- Appendix:
  - How are Stocks Traded?
  - Is Growth Always Good?
  - Growth Opportunities and P/E Ratios
  - Multi-Stage Growth

# Objectives

- Apply present and future value concepts, with uncertainty, to more complex valuation problems, specifically **stocks**.
- Explore the Gordon Growth Model.
  - Is it useful in real life? How do financial analysts use it?

# Returns from Investment in Stocks

***Main Principle: "Asset Prices Are Present Values"***

# Returns from Investment in Stocks

- The idea that the market price is the present value of expected future *cash flows* applies to all financial contracts.
  - Adjusting for risk is the tricky part!

- The value of the asset is the value today of the future cash flows.
  - This is a present value calculation!

- Owning **stock** gives you:
  - A claim to future dividend payments **and** capital gains or losses.
  - Cash from stock repurchases.
  - Voting rights.
    - Vote in elections for appointments to the board of directors.

# Returns from Investment in Stocks

- **Stock:** Common stockholders are the owners of a company. A share of common stock gives its holder claims to the cash flows of a company after the company has met its fixed obligations.

- The difference between the cash flows of common stock and bonds is that equity holders get paid last while debt holders are paid first.
  - Equity holders are known as **residual claim holders**.

- When an investor purchases a share of stock, she benefits in **two** ways:
  - **Capital Gain:** The price at which the investor can sell the stock might be higher than the price at which she purchased the stock.
  - **Dividend Yield:** Firm announces dividend payments.

# How do Investors Set the Price of a Stock?

# How do Investors Set the Price of a Stock: $P_0$

- Suppose you buy a share of stock for $P_0$:
  - You expect to be able to sell it next year for $P_1$
  - You expect end-of-year dividend $Div_1$

- The total return has **two** components:
  - **Capital Gain Return** = $(P_1 – P_0)/ P_0$
  - **Dividend Yield** = $Div_1/P_0$ [Return From Dividends]

- If the investors can lend money at a rate of $r$ in an alternative opportunity that is similar in risk to the stock, then the return on the stock must equal the alternative return:

$$\boldsymbol{r} = \frac{Div_1}{P_0} + \left(\frac{P_1 - P_0}{P_0}\right) \rightarrow P_0 = \frac{P_1 + Div_1}{1 + \boldsymbol{r}}$$

- **Remark:** r is also known as the *market capitalization rate*.

# How do Investors Set the Price of a Stock: $P_0$

## Example: Apple (AAPL) Forecasts

- At the end of September 2018, Apple's share price was $P_0 = \$225.74$ per share (Yahoo Finance).
- The Institutional Brokers' Estimate System (IBES) is a database that compiles the different estimates made by stock analysts. It is available through Thomson One.
  - Forecast a dividend per share of $3.13 over the next year (ie: FY-Sep-19).
  - The median price forecast is $243.39 per share.

APPLE INC. (Non Per Share in USD MM, Per Share in USD)

**Important Notices**

25-Oct-16 SEP19 Majority Estimates reflect adoption of FAS123(R)

| | Mean | FY-Sep.17 | | CURRENT FY-Sep.18 | FY-Sep.19 |
|---|---|---|---|---|---|
| **Income Statement** | | | | | |
| Revenue | 227,368.31 | 229,234.00 | A | 263,951.12 | 281,296.65 |
| Gross Margin (%) | 38.47 | 38.47 | A | 38.37 | 38.92 |
| EBIT | 60,895.33 | 61,344.00 | A | 70,302.86 | 74,452.26 |
| EBITDA | 71,285.26 | 71,501.00 | A | 81,444.05 | 85,899.80 |
| Pre-tax Profit | 63,335.56 | 64,089.00 | A | 72,243.45 | 75,562.87 |
| Net Income | 47,327.96 | 48,351.00 | A | 58,872.04 | 63,647.49 |
| Reported Net Profit | 47,349.08 | 48,351.00 | A | 58,755.65 | 63,258.07 |
| Reported Pre-tax Profit | 63,203.32 | 64,089.00 | A | 71,997.44 | 74,758.85 |
| Per Share Data | | | | | |
| EPS (Options Expense Included) | 9.00 | 9.21 | A | 11.78 | 13.75 |
| EPS - Secondary Mean (Options Expense Excluded) | NA | NA | | 11.74 | 13.39 |
| EPS - Fully Reported | 9.01 | 9.21 | A | 11.74 | 13.64 |
| EBITDA per Share | 13.58 | 13.62 | A | 16.38 | 17.87 |
| Dividend per Share | 2.41 | 2.40 | A | 2.76 | 3.13 |

**Target Price and Long Term Growth**

| | Target Price(USD) | LTG(%) |
|---|---|---|
| Mean | 238.93 | 11.67 |
| Median | 243.39 | 11.00 |
| High | 310.00 | 13.00 |
| Low | 175.00 | 11.00 |
| Standard Dev | 27.78 | 1.15 |
| Total# | 36 | 3 |

# How do Investors Set the Price of a Stock: $P_0$

**Example: Price Today**

- At the end of September 2018, Apple's share price was $P_0$ = \$225.74 per share. Investors expect a $Div_1$ = \$3.13 cash dividend over the next year. They also expect the stock to sell in a year for $P_1$ = \$243.39.

$$\boldsymbol{r} = \frac{Div_1}{P_0} + \left(\frac{P_1 - P_0}{P_0}\right) \rightarrow \boldsymbol{r} = \frac{3.13}{225.74} + \left(\frac{243.39 - 225.74}{225.74}\right) = 0.092 = 9.2\%$$

- If you are given investors' forecasts of dividend and price *and* the expected return offered by other equally risky stocks ***r***, you can predict today's price.

$$P_0 = \frac{P_1 + Div_1}{1 + \boldsymbol{r}} = \frac{243.39 + 3.13}{1 + 0.092} = 225.74$$

- **Bottom Line:** The price today is simply the PV of the expected payoff in a year, discounted at r.

# How do Investors Set the Price of a Stock: $P_0$

- **Question:** Why is $P_0$ = 225.74 the correct price? At r = 9.2%, there is no other $P_0$ that could survive in competitive markets. Why?
  - If $P_0$ = \$230, then you could expect a return of:

$$\boldsymbol{r} = \frac{Div_1}{P_0} + \left(\frac{P_1 - P_0}{P_0}\right) \rightarrow \boldsymbol{r} = \frac{3.13}{230} + \left(\frac{243.39 - 230}{230}\right) = 0.072 = 7.2\%$$

  If $P_0$ > 225.74, the expected return r is **lower** than other securities of equivalent risk. Investors would shift their capital to other securities, forcing the stock price **down**!

  - If $P_0$ = \$220, then you could expect a return of:

$$\boldsymbol{r} = \frac{Div_1}{P_0} + \left(\frac{P_1 - P_0}{P_0}\right) \rightarrow \boldsymbol{r} = \frac{3.13}{220} + \left(\frac{243.39 - 220}{220}\right) = 0.121 = 12.1\%$$

  If $P_0$ < 225.74, the expected return r is **higher** than other comparable securities. Investors would rush to buy, forcing the price **up**!

# How do Investors Set the Price of a Stock: $P_0$

- **Basic Principle**
  - All securities in an **equivalent** risk class are priced to offer the same expected rate of return.
  - The price of a stock is set so that the stock's expected return equals that of the other stock with similar risk.

- **Implication:**
  - Price = PV of expected future dividends.

- **Remark:** So far we have ignored the question of where investors get their estimate of $P_1$. We need to start thinking about this issue.

# How do Investors Set the Price of a Stock: $P_1$

- The price investors are willing to pay today:

$$P_0 = \frac{Div_1 + P_1}{1 + r}$$

- Using the same formula, the price investors will be willing to pay at t=1 (assume that r remains constant over time) is

$$P_1 = \frac{Div_2 + P_2}{1 + r}$$

- So at t = 0, investors will be willing to pay:

$$P_0 = \frac{Div_1}{1 + r} + \frac{Div_2 + P_2}{(1 + r)(1 + r)}$$

- Substituting $P_2$, then $P_3$, then $P_4$,..., etc., we obtain

$$P_0 = \frac{Div_1}{1 + r} + \frac{Div_2}{(1 + r)^2} + \frac{Div_3}{(1 + r)^3} \ldots . \rightarrow P_0 = \sum_{i=1}^{\infty} \frac{Div_i}{(1 + r)^i}$$

# Investor Price Setting: Final Remarks

- The value of a firm's stock is the present value of **expected** future dividends!
  - The challenge is in estimating expected future dividends!
- If we look out far enough, the PV of the future dividends that the firm's stock price represents is **very little** (when discounted back to today) in comparison with the PV of the dividends we will receive before then.
- **Bottom Line:** This is how you should value a stock, regardless of how long you intend to hold it.
  - If you sell the share, the purchaser buys it for the PV of its future dividends from then on.

# The Gordon Growth Model

# The Gordon Growth Model (GGM)

- Assume:
  - The firm is expected to live forever.
  - Dividends grow at a constant rate **g** forever.
  - Dividends are paid yearly.
  - Market capitalization rate will remain constant at **r** where **r** > **g**.
    - Some practitioners call r the **Expected Return On Equity**.
  - The firm invests out of retained earnings only (internal financing)
- Under these (restrictive!) assumptions, the dividend stream is a **growing perpetuity** and the stock price is:

$$P_0 = \frac{Div_1}{r-g}$$

- Rearranging the pricing equation to get the return equation:

$$r = \frac{Div_1}{P_0} + g$$

# The Gordon Growth Model

- **Question:** How do we estimate this return equation in real life?
  - How do financial analysts use this equation?
  - Is it useful? **Yes!**

# Using The GGM To Set Prices

- Prices charged by **local electric** and **gas utilities** are regulated by state commissions.
  - Regulators try to keep consumer prices down but are supposed to allow the utilities to earn a fair rate of return.
    - The rate offered by securities that have the same risk as the utility's common stock.

- Interpretation from the US Supreme Court's Directive in 1944 [Federal Power Commission vs. Hope Natural Gas Company]:

*"The returns to the equity owner [of a regulated business] should be commensurate with returns on investments in other enterprises having corresponding risks"*

- **Bottom Line:** Utilities are mature, stable companies that offer a good example of how the GGM is applied in practice by financial analysts.

# Using The GGM To Set Prices

- Suppose you want to estimate the cost of equity for Duke Energy, a regulated electricity public utility that primarily serves North Carolina and South Carolina.
  - On October 19, 2018, its stock was selling at $82.75.
  - Dividend payments next year (and forever?) were expected to be $3.80 p/share.

$$\text{Dividend Yield} = \frac{Div_1}{P_0} = \frac{\$3.80}{\$82.75} = 0.046 = 4.6\%$$

- The **difficult** part is estimating g, the expected rate of dividend growth.
  - There are **two** possible options:
    - **Option A:** Use financial analysts forecasts.
    - **Option B:** Use Earnings Per Share (EPS).

# Duke Energy (DUK)

**DUKE ENERGY CORP (Non Per Share in USD MM, Per Share in USD)**

**Important Notices**

16-Feb-17 DEC19 Majority Estimates reflect adoption of FAS123(R)

| | Mean | FY-Dec.17 | | CURRENT FY-Dec.18 | FY-Dec.19 | FY-Dec.20 | FY-Dec.21 |
|---|---|---|---|---|---|---|---|
| **Income Statement** | | | | | | | |
| Revenue | 24,061.11 | 23,565.00 | A | 24,070.12 | 24,678.26 | 25,059.84 | 24,244.54 |
| Gross Margin (%) | 70.26 | 72.93 | A | 69.15 | 69.63 | 69.60 | NA |
| EBIT | 6,215.90 | 5,932.00 | A | 5,735.46 | 6,076.63 | 6,297.77 | 5,705.42 |
| Operating Profit | NA | 6,302.40 | | 5,513.20 | 6,082.21 | NA | NA |
| EBITDA | 9,693.53 | 9,337.00 | A | 9,562.49 | 10,072.93 | 10,420.01 | 9,809.11 |
| Pre-tax Profit | 4,628.66 | 4,502.00 | A | 4,023.45 | 4,358.14 | 4,588.53 | NA |
| Net Income | 3,163.62 | 3,199.00 | A | 3,283.57 | 3,608.41 | 3,818.19 | 4,068.22 |
| Reported Net Profit | 3,076.14 | 3,059.00 | A | 3,283.68 | 3,607.22 | 3,824.92 | NA |
| Reported Pre-tax Profit | 4,514.52 | 4,266.00 | A | 4,038.50 | 4,386.41 | 4,600.83 | 4,833.00 |
| Per Share Data | | | | | | | |
| EPS | 4.56 | 4.57 | A | 4.72 | 4.96 | 5.22 | 5.54 |
| Funds From Operations | NA | NA | | NA | 11.57 | 11.63 | 11.84 |
| EPS - Fully Reported | 4.38 | 4.36 | A | 4.56 | 4.96 | 5.21 | NA |
| EBITDA per Share | 13.88 | 13.34 | A | 13.49 | 14.10 | 14.68 | NA |
| Dividend per Share | 3.51 | 3.49 | A | 3.65 | 3.80 | 3.98 | 4.12 |
| **Cash Flow** | | | | | | | |
| Capital Expenditure | 9,095.00 | 8,052.00 | A | 10,673.38 | 10,719.33 | 9,093.75 | 8,400.00 |
| Per Share Data | | | | | | | |
| Cash Flow per Share | 9.81 | 9.48 | A | 10.11 | 11.26 | 11.63 | 11.84 |
| Free Cash Flow per Share | NA | NA | | -5.73 | -3.48 | -0.74 | 0.41 |
| **Balance Sheet** | | | | | | | |
| Net Asset Value | 41,466.67 | 41,737.00 | A | 44,170.67 | 45,364.00 | 46,758.50 | 48,381.00 |
| Net Debt | 54,073.30 | 54,084.00 | A | 57,592.65 | 62,135.22 | 64,140.07 | NA |
| Per Share Data | | | | | | | |
| Book Value per Share | 59.11 | 59.63 | A | 61.52 | 62.46 | 63.93 | NA |
| Tangible Book Value per Share | 31.92 | 31.92 | A | 35.06 | 36.00 | 37.46 | NA |
| **Valuation** | | | | | | | |
| ROA (%) | 2.85 | 2.36 | A | 2.95 | 2.98 | 3.13 | NA |
| ROE (%) | 7.72 | 7.73 | A | 7.81 | 8.07 | 8.28 | NA |

**Recommendations**

| Recommendations | # of Brokers Current | Month Ago 1 | 2 | 3 |
|---|---|---|---|---|
| Strong Buy | 2 | 2 | 2 | 2 |
| Buy | 5 | 5 | 5 | 4 |
| Hold | 10 | 11 | 11 | 9 |
| Underperform | 1 | 1 | 1 | 2 |
| Sell | 0 | 0 | 0 | 1 |
| Total | 18 | 19 | 19 | 18 |

**P/E Ratios** More Ratios

| Period | P/E Ratio | PEG Ratio |
|---|---|---|
| FY0 | 18.107 | NA |
| FY1 | 17.521 | 5.237 |
| FY2 | 16.668 | 3.255 |
| FY3 | 15.840 | 3.030 |

**Target Price and Long Term Growth**

| | Target Price(USD) | LTG(%) |
|---|---|---|
| Mean | 84.94 | 4.40 |
| Median | 85.00 | 4.40 |
| High | 92.00 | 4.73 |
| Low | 72.00 | 4.06 |
| Standard Dev | 4.42 | 0.47 |
| Total# | 17 | 2 |

Sell Strong Buy

**Current Rec. Mean: 2.56**

**Thomson One, accessed October 21, 2018**

# Using The GGM To Set Prices

- **Option A:** Consult the view of security analysts who study the prospects of each company. Use Thomson One!
  - They often forecast growth rates over the next 5 years!
  - These estimates may provide an indication of expected long-run growth path.
  - For Duke Energy, they were forecasting an annual growth of **4.4%.**

- This, together with the dividend yield, gives an estimate of the cost of capital:

$$r = \frac{Div_1}{P_0} + g = 4.6\% + 4.4\% = 9.0\%$$

- In this calculation, we are assuming that **earnings** and **dividends** are forecasted to grow forever at the same rate, g.

# Using The GGM To Set Prices

- **Option B:** Use Earnings Per Share. We need to introduce some useful definitions:
  - **Earnings Per Share** (EPS) refers to the total cash flows that are available to be paid out as dividends or to be reinvested in the firm. We will compute it at Net $\text{Income}_t$ /Total # $\text{Shares}_t$.
  - **Payout Ratio** is the fraction of earnings paid out in the form of dividends.
    - **$\text{Payout Ratio}_t = \text{Div}_t/\text{EPS}_t$**
  - **Plowback Ratio** is the fraction of earnings that the company plows back into the business. Fraction of earnings reinvested.
    - **$\text{Plowback Ratio}_t = 1 – \text{Payout Ratio}_t = b_t$**
  - **Return on Equity (ROE)** is the cash flow generated by capital as a percentage of Book Equity Per Share (BE).
    - **$\text{ROE}_t = \text{EPS}_t/(\text{Book Equity Per Share}_{t-1})$**

# Using The GGM To Set Prices

- For Duke:
  - $Div_t/EPS_t = 3.65/4.72 = 77\%$
  - Plowback Ratio = $1 - Div_t/EPS_t = 23\%$
  - $ROE_t = EPS_t/\text{Book Equity Per Share}_{t-1} = 4.72/59.63 = 7.9\%$
- The growth rates of book equity, earnings, and dividends are **all equal** and given by:

$$g = \text{Plowback} \times \text{ROE (Why?)}$$

$$g = 0.23 \times 0.079 = 0.018 = 1.8\%$$

- This, together with the dividend yield, gives the **second** estimate of the cost of capital:

$$r = \frac{Div_1}{P_0} + g = 4.6\% + 1.8\% = 6.4\%$$

# Some Important Remarks About GGM

- Although both methods seem reasonable enough, there are obvious **dangers** in analyzing any single firm's stock!
  - Regular future growth is at best an approximation.
  - Any estimate of r for a **single** common stock is "noisy" and subject to error.

- Good practice does not put too much weight on single-company cost of equity estimates.
  - Collect sample of similar companies, estimate r for each company, and take an **average** → Use a large sample.

- Avoid using the formula for firms having high current rates of growth.
  - That growth **cannot** be sustained indefinitely!

# Some Important Remarks About GGM

- Our two cost of capital estimates for Duke (9.0% and 6.4%) are quite different.
- Let’s consider “comparable” firms, as informed by Thomson One, and compute a market cap weighted average.

Comparables | DUKE ENERGY CORPORATION (DUK-US)

Peers (By Criteria)

Market Data & Price Multiples | EV Multiples & Credit Ratios | Key Financials & Effectiveness

**Market Data & Price Multiples**

| Name | Ticker | Last Period End Date | Price | 52 Week Low | 52 Week High | Dividend Yield TTM | Total Return 1 Yr |
|---|---|---|---|---|---|---|---|
| **DUKE ENERGY CORPORATION** | **DUK-US** | **06/30/2018** | **82.75** | **71.96** | **91.80** | **4.64%** | **(1.14%)** |
| THE SOUTHERN COMPANY | SO-N | 06/30/2018 | 45.07 | 42.38 | 53.51 | 5.51% | 1.61% |
| EXELON CORPORATION | EXC-N | 06/30/2018 | 44.13 | 35.57 | 45.05 | 3.22% | 21.83% |
| AMERICAN ELECTRIC POWER COMPANY, INC. | AEP-N | 06/30/2018 | 73.26 | 62.71 | 78.07 | 3.50% | 5.78% |
| CENTERPOINT ENERGY, INC. | CNP-N | 06/30/2018 | 28.12 | 24.81 | 30.17 | 4.06% | 5.19% |
| NEXTERA ENERGY, INC. | NEE-N | 06/30/2018 | 173.46 | 145.10 | 175.65 | 2.62% | 22.18% |
| FIRSTENERGY CORP. | FE-N | 06/30/2018 | 38.85 | 29.33 | 38.99 | 3.87% | 28.09% |
| CMS ENERGY CORPORATION | CMS-N | 06/30/2018 | 50.68 | 40.48 | 50.98 | 2.93% | 4.26% |
| EDISON INTERNATIONAL | EIX-N | 06/30/2018 | 70.31 | 57.63 | 83.38 | 3.53% | (16.07%) |
| CONSOLIDATED EDISON, INC. | ED-N | 06/30/2018 | 77.38 | 71.12 | 89.70 | 3.80% | (0.04%) |
| Mean | | | - | - | - | 3.82% | 7.41% |
| Median | | | - | - | - | 3.80% | 5.19% |
| High | | | 173.46 | 145.10 | 175.65 | 5.51% | 28.09% |
| Low | | | 28.12 | 24.81 | 30.17 | 2.62% | (16.07%) |

# Some Important Remarks About GGM

- The market cap weighted averages (9.92% and 7.78%) are closer together.
  - As a next step, we could further refine our selection of "comparable" firms, possibly removing the firms with relatively high long-term growth forecasts.
  - Ultimately, we may wish to argue that one method is preferred to the other (for our specific case) or use the average.

| Name | Dividend Yield | LT Growth Forecast | Div/sh (2018) | EPS (2018) | BE/sh (2017) | Plowback Ratio | ROE | Cost of Capital (Method A) | Cost of Capital (Method B) | Market Cap ($B) |
|---|---|---|---|---|---|---|---|---|---|---|
| Duke Energy | 4.64% | 4.40% | 3.65 | 4.72 | 59.63 | 23% | 7.92% | 9.04% | 6.43% | 58947 |
| The Southern Company | 5.51% | 1.70% | 2.37 | 3.00 | 24.51 | 21% | 12.24% | 7.21% | 8.08% | 45707 |
| Exelon Corp. | 3.22% | 6.12% | 1.38 | 3.12 | 31.00 | 56% | 10.06% | 9.34% | 8.83% | 42625 |
| American Electric Power | 3.50% | 5.73% | 2.51 | 3.94 | 37.19 | 36% | 10.59% | 9.23% | 7.35% | 36112 |
| Centerpoint Energy | 4.06% | 8.70% | 1.12 | 1.58 | 10.88 | 29% | 14.52% | 12.76% | 8.29% | 14958 |
| Nextera Energy | 2.62% | 9.53% | 4.44 | 7.75 | 60.02 | 43% | 12.91% | 12.15% | 8.13% | 81805 |
| **Market Cap Weighted Average** | | | | | | | | **9.92%** | **7.78%** | |

# Some Important Remarks About GGM

| GGM Cost of equity Estimates For US Railroads, 2014 | | | |
|---|---|---|---|
| Public Traded Firm | Average Dividend Yield | Forecasted Growth Rate | Cost of Equity |
| Burlington Northern Santa Fe | 1.85% | 9.12% | 10.97% |
| CSX | 1.29% | 11.37% | 12.66% |
| Norfolk Southern | 1.27% | 11.79% | 13.06% |
| Union Pacific | 1.44% | 12.05% | 13.49% |
| **Market Value Weighted Average** | 1.48% | 11.13% | 12.61% |

- Such high forecasted growth rates can rarely be sustained **indefinitely**, but the constant GGM assumes it can!
  - This will lead to an **overestimate** of r!
- In general, we can think of stock price as the capitalized value of average earnings under a no growth policy, **plus** PV(GO), the present value of growth opportunities.

$$P_0 = \frac{EPS_1}{r} + PV(GO)$$

# Takeaways

- The idea that value is based on the present value of future cash-flows applies to all financial contracts, including stocks.
- The value of a firm’s stock is the present value of expected future dividends!
- The Gordon Growth Model (GGM) is a relatively simple way to evaluate stock prices in a basic and intuitive way.
  - The return can be decomposed into two components:
    - Dividend Yield.
    - Capital gain
  - The price of a firm's stock can be divided up into:

$$P_0 = \frac{EPS_1}{r} + PV(GO)$$

# Appendix: How are Common Stocks Traded?

# How are Common Stocks Traded?

- The sale of **new** shares to raise new capital occurs in the **primary market**
- Most trades of shares take place in **existing** shares, which investors buy from each other. This is known as the **secondary market** for shares.
  - These trades do not raise new capital for the firm.
  - The principal secondary market for shares is the New York Stock Exchange (NYSE).
- The NYSE is an **auction market** in which a specialist (person responsible for trading in each stock) acts as an auctioneer, matching up would-be buyers and sellers.
- The NYSE is not the only market in the United States.

# How are Common Stocks Traded?

- Many stocks trade *over the counter* by a network of dealers, who display the prices at which they are prepared to trade on a system of computer terminals known as Nasdaq (National Association of Securities Dealers Automated Quotations System).
  - If you like the price that you see on the Nasdaq screen, you simply strike a bargain with the dealer.
  - Nasdaq is the largest electronic screen-based equity securities market in the United States.

- Nasdaq is an example of a **dealer market**, in which all trades take place between the investor and one of a group of dealers.
  - Dealer markets are rare for equity trading.
  - Bonds are commonly traded in a dealer market.

# How are Common Stocks Traded?

- The prices at which stocks are traded are summarized in the daily press.
  - *The Wall Street Journal* records the day's trading in Walt Disney (DIS) in the following way on October 19, 2018:

| YTD | 52 Weeks | | | | | | Vol | | Net |
|---|---|---|---|---|---|---|---|---|---|
| % Chg | Hi | Lo | Stock (SYM) | Div | Yld% | PE | 1000s | Close | Chg |
| +5.4 | 119.17 | 96.89 | Walt Disney (DIS) | 1.68 | 1.4 | 15 | 10006 | 118.90 | 2.72 |

- This information summarizes:
  - Total Volume Traded = 10,006 × 1,000 = 10,006,000 shares
  - Close of day stock price at $118.90 p/share.
    - Up $2.72 from the day before.
  - Stock increased by 5.4% from the start of 2018.
  - Number of total shares outstanding: 1.5B shares.
  - Market Value Capitalization = 1.5B × $118.90 = **$178B!**

# How are Common Stocks Traded?

- It also provides three other factors about DIS's stock:
  - DIS pays an annual dividend of $1.68 a share.
  - The dividend yield on the stock is 1.4%
  - Price/Earnings Ratio (P/E) = 15

# Appendix: Is Growth Always Good?

# Is Growth Always Good?

- The payout ratio is a **decision variable**!
  - You should choose it to make the price of the stock as big as possible.
  - A lower payout ratio leads to higher growth g, but is growth always good? **No!**
    - You need to ask: at what rate are you plowing back?

- In general, we can think of stock price as the capitalized value of average earnings under a no growth policy, **plus** PV(GO), the present value of growth opportunities.

$$P_0 = \frac{EPS_1}{r} + PV(GO)$$

- Let's see some examples…

# Is Growth Always Good? Application

- Assume a firm with:
  - ROE = 8% on existing assets (expected to continue indefinitely).
  - Book value of assets of $50 per share ($BE_0$)
  - A market capitalization rate of 10% (r).
  - A payout ratio of 100% (i.e. plowback ratio = 0).
- Answer the following questions:
  a. What is the share price with **no** plowback?
  b. Suppose that, instead, the plowback ratio is set at **60%.** What happens to the share price?

# Is Growth Always Good? Application (cont.)

## Share Price With No Plowback

- The existing EPS is $0.08 \times \$50 = \$4$.

- Using the GGM:
  - Because there is no plowback, all earnings are paid out as dividends → $g = 0$
  - $P_0 = Div_1/r = \$4/0.10 = \$40.00$

- The **earnings-price ratio** is $EPS_1/P_0 = \$4/\$40 = 0.10 = r$

- **Bottom Line:** This ratio equals the market capitalization rate because the $4 dividend is the only return on holding the stock.

# Is Growth Always Good? Application (cont.)

## Plowback At 60%

- Using:

$$Div_1 = \text{Payout Ratio} \times EPS_1 = \text{Payout Ratio} \times ROE_1 \times BE_0$$

$$Div_1 = 0.40 \times 0.08 \times \$50 = \$1.6$$

- The dividend growth rate will be:

$$g = \text{Plowback Ratio} \times ROE$$

$$g = 0.60 \times 0.08 = 4.8\%$$

- Using the Gordon model, the stock price will be:

$$P_0 = Div_1 / (r - g) = \$1.6/(0.10 - 0.048) = \mathbf{\$30.77}$$

- **Bottom Line:** The value of the firm has gone **down** because of the plowing back of earnings into the firm! Why?

# Is Growth Always Good? Application (cont.)

- The reason is the plowed back earnings earn the current ROE of 8%, while the market capitalization rate is 10%!
  - This is an example where growth is not good!

- A firm has no business retaining earnings if it cannot invest at a rate **higher** than that which investors could earn in an alternative investment of equivalent risk.

- **Question:** How would your results change if the ROE = 12% and the market capitalization rate is still 10%?

# Is Growth Always Good? Application (cont.)

## Plowback at ROE > r

- With no plowback:

$$P_0 = EPS/r = [(\text{Book Value p/Share}) \times (\text{ROE})]/r$$
$$= \$50 \times 0.12 / 0.10 = \$60$$

- With plowback at 60%:

$$\text{Div}/(\text{Book Equity Per Share}) = \text{Payout Ratio} \times \text{ROE}$$
$$\text{Div}_1 = 0.40 \times 0.12 \times \$50 = \$2.4$$

- The dividend growth rate is:

$$g = \text{Plowback Ratio} \times \text{ROE} = 0.60 \times 0.12 = 7.2\%$$

- Using the Gordon Model, the stock price is:

$$P_0 = \text{Div}_1 / (r - g) = \$2.4/(0.10 - 0.072) = \mathbf{\$85.71}$$

$85.71 – $60.00 = **$25.71** = PV(GO) → **Growth Opportunities!**

# Calculating P/E Ratios

- People often look at P/E ratios of firms.
- In the first case, the firm has no growth opportunities.
  - The P/E equals

$$EPS_1 / P_0 = \$4/\$40 = 0.10 = r \rightarrow \textbf{P/E = 10}$$

  - Since there are no growth opportunities, the earnings should all be paid out as dividends and the dividends provide the only return → **Low P/E firm!**

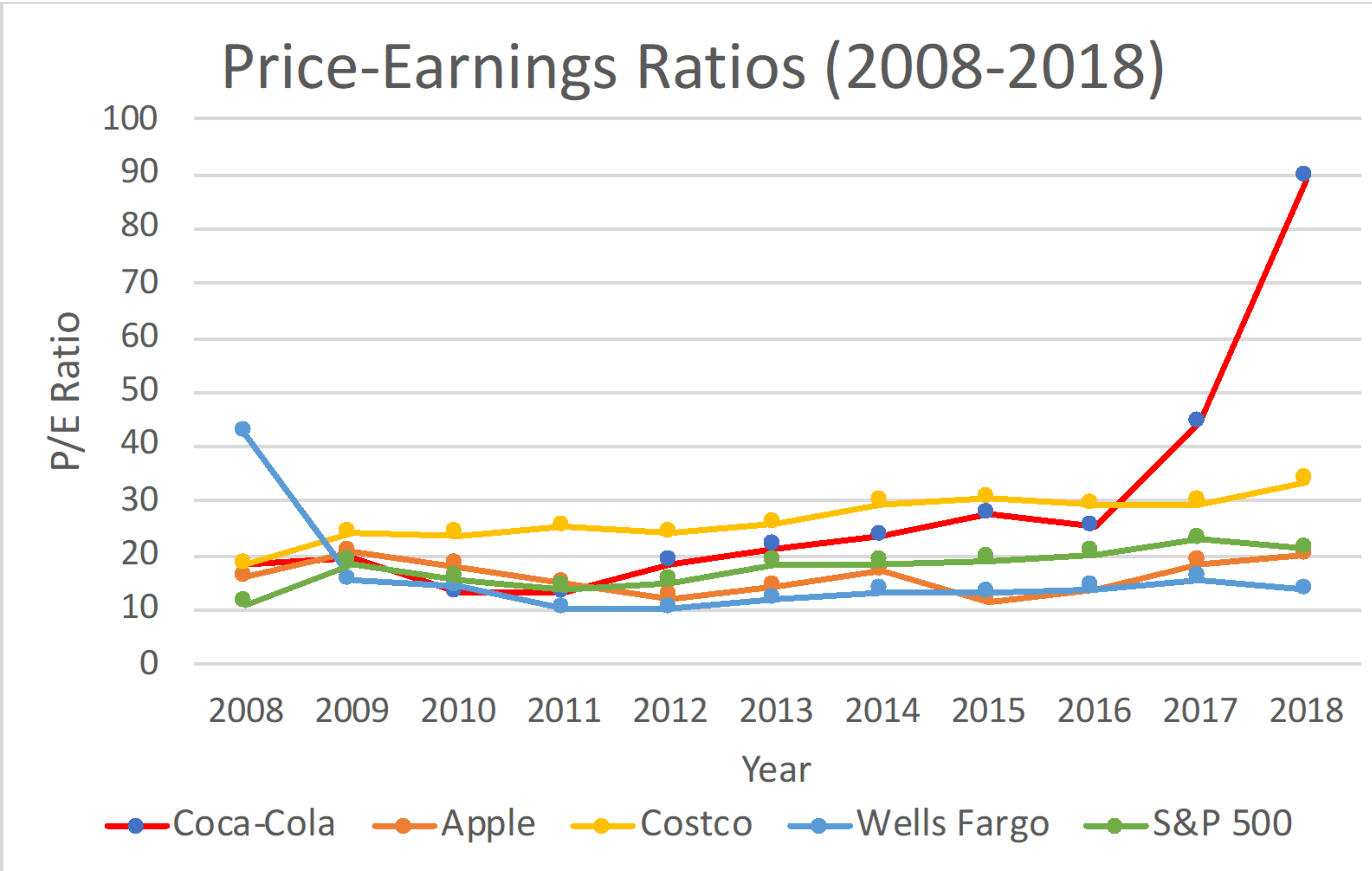
- In the second case, the firm has valuable growth opportunities.
  - The P/E equals

$$EPS_1 / P_0 = \$6/\$85.71 = 7\% < 10\% \rightarrow \textbf{P/E = 14.29}$$

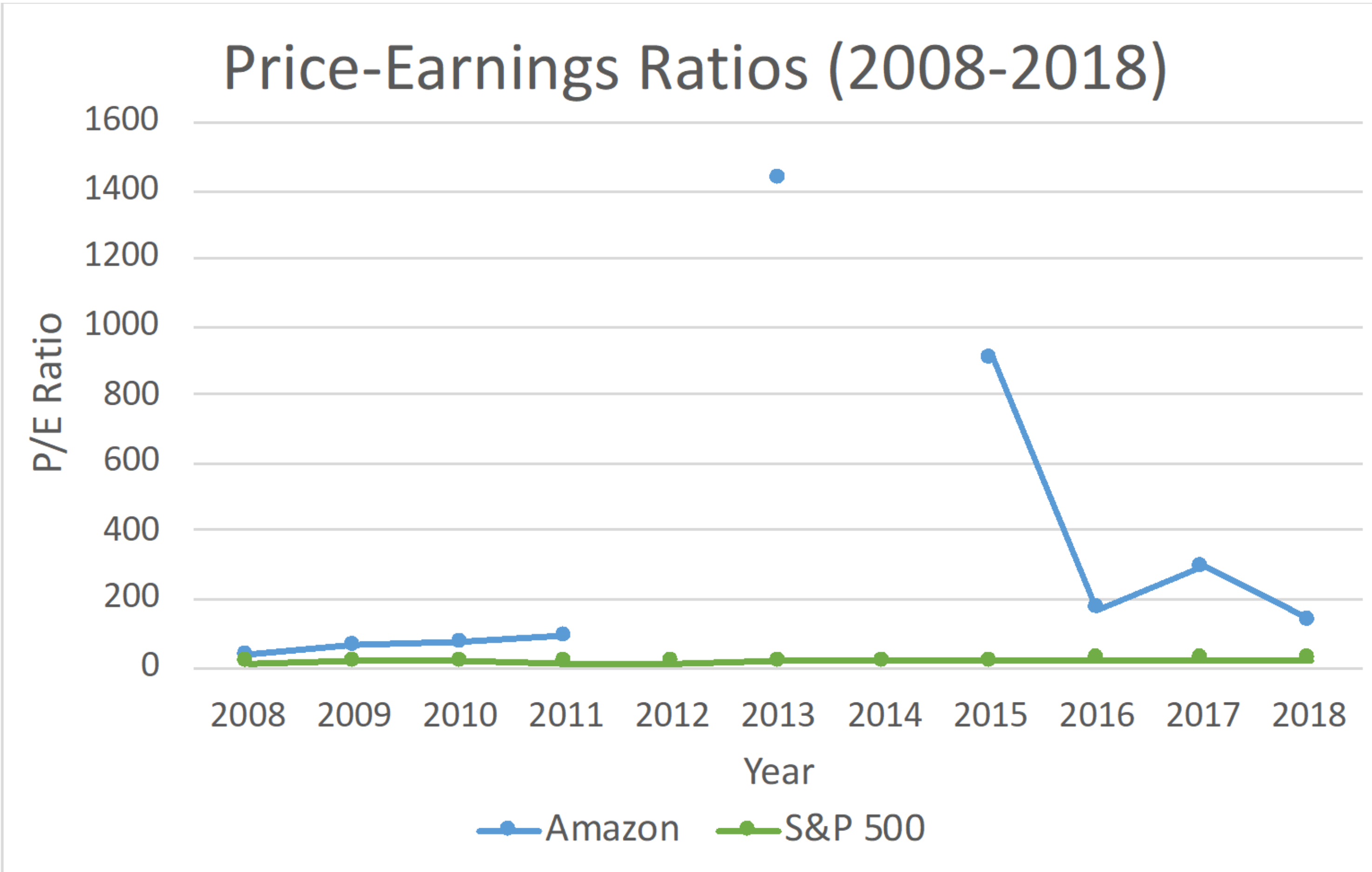
  - Earnings do not account for the growth opportunities.
    - There will be capital gains! → **High P/E firm!**

- **Bottom Line:** Growing firms have high P/E ratios!

# Historical P/E Ratios



# Historical P/E Ratios




- **Question**: Why is there missing data for Amazon (eg: 2012)?

# A Case When Growth Is Good…

### Cost of capital

Does the prospect of tightening by the US Federal Reserve dull the animal spirits of dealmakers? One might have thought so. Rising interest rates should increase companies' cost of capital, raise the hurdle rate for investments and push up funding costs. But previous rate rise cycles reveal a more nuanced picture. A Citigroup study suggests that companies increased capital expenditure substantially in the last three Fed tightening episodes (1988-89, 1994-95 and 1999-2000) and increased merger and acquisition activity in the previous two cycles.

One explanation is that the prospect of increased growth can more than offset higher capital costs. Companies were not deterred from borrowing since many hoped that improved cash flows would offset the threat to credit quality.

Where does that leave companies this time? Given the extremely benign tightening and the cash sitting on corporate balance sheets, M&A activity has been surprisingly listless. That may be because boardrooms are not looking at the cost of debt. They are looking at risk and they do not like what they see. The equity risk premium, the extra return to compensate investors for market risk, is at historically high levels. Lehman Brothers calculates the premium is roughly 5.5 per cent, close to the 90th percentile, based on data going back to 1982. Chief executives are a nervous lot at the moment, but it is not interest rates that are unnerving them.

**Wall Street Journal, September 2005.**

**Increasing growth opportunities can more than offset higher costs of capital.**

**Many companies were not deterred from borrowing since many hoped that better cash flows would offset the threat to credit quality.**

# A Case When Growth Is Doubtful…

**Comcast Slides After $39 Billion Sky Takeover Spooks Investors (Bloomberg, September 23, 2018)**

- The acquisition allows the cable giant (Comcast) to double its number of TV subscribers.
- The deal was announced in September 2018.
  - Price tag = $39B
  - At the announcement of the deal, Comcast stock declined by **8.1%!**
- **Question:** Is the market seeing the Present Value of Growth Opportunities as **negative**?

# Appendix: Growth Opportunities and P/E Ratios

# Growth Opportunities and P/E Ratios

- Investors often use the terms **growth stocks** and **income stocks**.
  - Growth stocks are linked to the expectation of capital gains.
  - Income stocks are linked to the expectation of cash dividends.

- In the setting of the Gordon Growth Model, the **Price/Earnings Ratio (P/E Ratio)** is:

$$P_0 = \frac{Div_1}{r-g} = \frac{Payout\ Ratio\ \times EPS_1}{r - Plowback\ Ratio \times ROE}$$
$$\rightarrow \frac{P_0}{EPS_1} = \frac{Payout\ Ratio}{r - Plowback\ Ratio \times ROE}$$

  - A Growth Stock is a stock with a **high** $P_0/EPS_1$
  - A Income Stock is a stock with a **low** $P_0/EPS_1$

# Growth Opportunities and P/E Ratios (cont.)

- **Remark I:** A **high** ROE implies a high growth rate and a **high** P/E ratio.
- **Remark II:** But note that you will have **high** growth rate but a **lower** P/E ratio if $g$ is high due to a high Plowback Ratio and not a high ROE!
  - Why? Because you can plowback and still have ROE < r!
- Let's understand why the GGM can be expressed as:

$$P_0 = \frac{EPS_1}{r} + PV(GO)$$

- **Question:** How much of the stock price is **due** to growth opportunities?

# Growth Opportunities and P/E Ratios (cont.)

- If the firm chose a Payout Ratio = 1 with g = 0, the price (and P/E Ratio) would be:

$$P_0 = \frac{Div_1}{r-g} = \frac{EPS_1}{r} \rightarrow \frac{P_0}{EPS_1} = \frac{1}{r}$$

- If ROE > r, the firm should plowback some earnings. Suppose the firm is able to plowback a fraction of earnings, $b$. Let's compute the P/E Ratio:

$$P_0 = \frac{EPS_1(1-b)}{r-g} \rightarrow \frac{P_0}{EPS_1} = \frac{1-b}{r-g}$$

  - Using g = ROE × b and manipulating we obtain:

$$\frac{P_0}{EPS_1} = \frac{1-b}{r - ROE \times b} = \frac{1}{r} + \frac{(ROE - r)b}{r(r - ROE \times b)} = \frac{1}{r} + PV(GO)$$

# Growth Opportunities and P/E Ratios (cont.)

- If b = 0:

$$\frac{P_0}{EPS_1} = \frac{1}{r}$$

- If b > 0 and ROE > r then,

$$\frac{(ROE - r)b}{r(r - ROE \times b)} > 0 \rightarrow PV(GO) > 0 \Rightarrow \frac{P_0}{EPS_1} > \frac{1}{r}$$

  - This represents good investment opportunities: **PV(GO) > 0!**
    - A dollar of earnings at date 1 is now **more** valuable than one dollar. Part of that dollar is re-invested at ROE > r.

- If b > 0 and ROE < r then,

$$\frac{(ROE - r)b}{r(r - ROE \times b)} < 0 \rightarrow PV(GO) < 0 \Rightarrow \frac{P_0}{EPS_1} < \frac{1}{r}$$

  - This represents bad investment opportunities: **PV(GO) < 0!**
    - A dollar of earnings at date 1 is worth **less** than one dollar. Part of that dollar is re-invested at ROE < r.

# If Profits Grow, How Can The Market Sink?

**New York Times, 2015**

- There has generally been an **inverse** relationship between overall earnings growth rates and the stock market's return.
  - Except during the market's worst periods.
  - The market has performed best during quarters when earnings are as much as 25% below earlier years.
- The problem is that such growth usually leads to **higher** interest rates.
  - When interest rates rise, the NPV of future earnings, cash flows, and dividends declines. This causes the market to decline!
- **Bottom Line:** The Fed is unlikely to feel pressure to raise rates when one company reports better-than-expected earnings, but feels the pressure when aggregate market earnings rise quickly.

# Estimating PV(Growth Opportunities)

- The first column shows the stock price for these companies.

| Estimating PV(Growth Opportunities) | | | | | |
|---|---|---|---|---|---|
| **Stock** | **Stock Price, $P_0$** | **EPS** | **Cost of Equity, r** | **PV(GO) = $P_0$ - EPS/r** | **PV(GO), Percent of Stock Price** |
| **Income stocks:** | | | | | |
| Citigroup | 48.09 | 3.82 | 0.12 | 15.44 | 32% |
| Dow Chemical | 39.69 | 2.25 | 0.08 | 12.25 | 31% |
| General Motors | 47.42 | 6.49 | 0.10 | -16.21 | -34% |
| J. C. Penney | 33.86 | 1.40 | 0.07 | 14.14 | 42% |
| **Growth stocks:** | | | | | |
| Amazon.com | 43.60 | 1.14 | 0.17 | 37.01 | 85% |
| Cisco Systems | 20.91 | 0.76 | 0.17 | 16.54 | 79% |
| Microsoft | 26.13 | 1.22 | 0.13 | 16.96 | 65% |
| Starbucks | 38.92 | 0.95 | 0.06 | 22.54 | 58% |

- **Questions:**
  - Where is the main source of value from growth stocks?
  - What about GM? What is the market telling us?

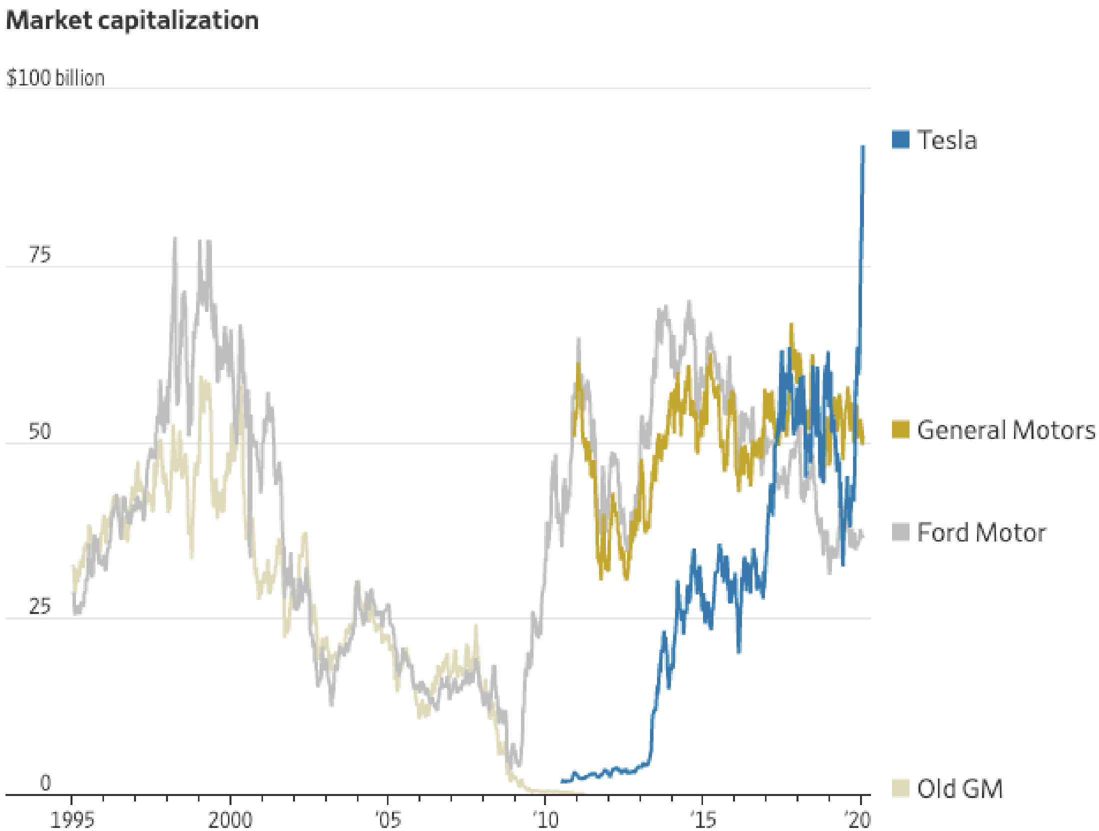
# Estimating PV(Growth Opportunities) (cont.)

- **Growth Stocks?**
  - Most of the growth rate comes from PV(GO).
    - Investors' expectations that the company will be able to earn more than the cost of capital on its future investments.
- **Income Stocks?**
  - Most of the growth is derived from the earning power of the existing assets.
- **General Motors?** → Two possible messages:
  - Investors expected GM to invest in negative-NPV projects.
  - We overestimated the earning power of GM's existing assets and attributed too much value to these assets and too little to PV(GO).

# Estimating PV(Growth Opportunities) (cont.)

## The Investor Clash Behind Tesla's Surge Toward $100 Billion in Market Value

Warring camps step up disagreement over what the shares are worth




- Tesla's market capitalization increased from $33.3 B in May 2019 to $92.0 B in Jan 2020 (+176%)
  - Why?

# Appendix: Multi-Stage Growth

# Multi-Stage Growth

- Suppose the conditions for the Gordon Growth Model are not met because growth will **change** over time.
  - We can apply a Multi-Stage Model.

- **Example:** Find the time 0 stock price of a small growing firm for which:
  - From year 1 through 12, ROE = 12% and b = 100%
  - From year 13 on, ROE = 10% and b = 40%.
  - $BE_0$ = $50
  - r = 10%

# Multi-Stage Growth: Example (cont.)

- **Step 1:** Work out book value of equity at time 12.

$$g_{0\text{-}12} = b \times ROE = 12\%$$

$$BE_{12} = BE_0 \times (1 + g)^{12} = \$50 \times (1.12)^{12} = \$194.80$$

- **Step 2:** Work out the stock price at time 12.

$$EPS_{13} = BE_{12} \times ROE_{13} = \$194.80 \times 10\% = \$19.48$$

$$Div_{13} = (1 - b) \times EPS_{13} = 0.6 \times \$19.48 = \$11.69$$

$$g_{13} = b \times ROE = 0.4 \times 10\% = 4\%$$

$$P_{12} = \frac{Div_{13}}{r - g_{13}} = \frac{\$11.69}{0.1 - 0.04} = \$194.80$$

- **Step 3:** Work out the stock price at time 0.

$$P_0 = \frac{P_{12}}{(1 + r)^{12}} = \frac{\$194.80}{(1.1)^{12}} = \$62.07$$

# Multi-Stage Growth: Example (cont.)

- Price Dynamics:
  - P grows by 10% for the first 12 years.
  - BE grows at 12%.
    - GGM does not apply here!
  - After t = 12, GGM applies since P and BE grow at the same rate of 4%.